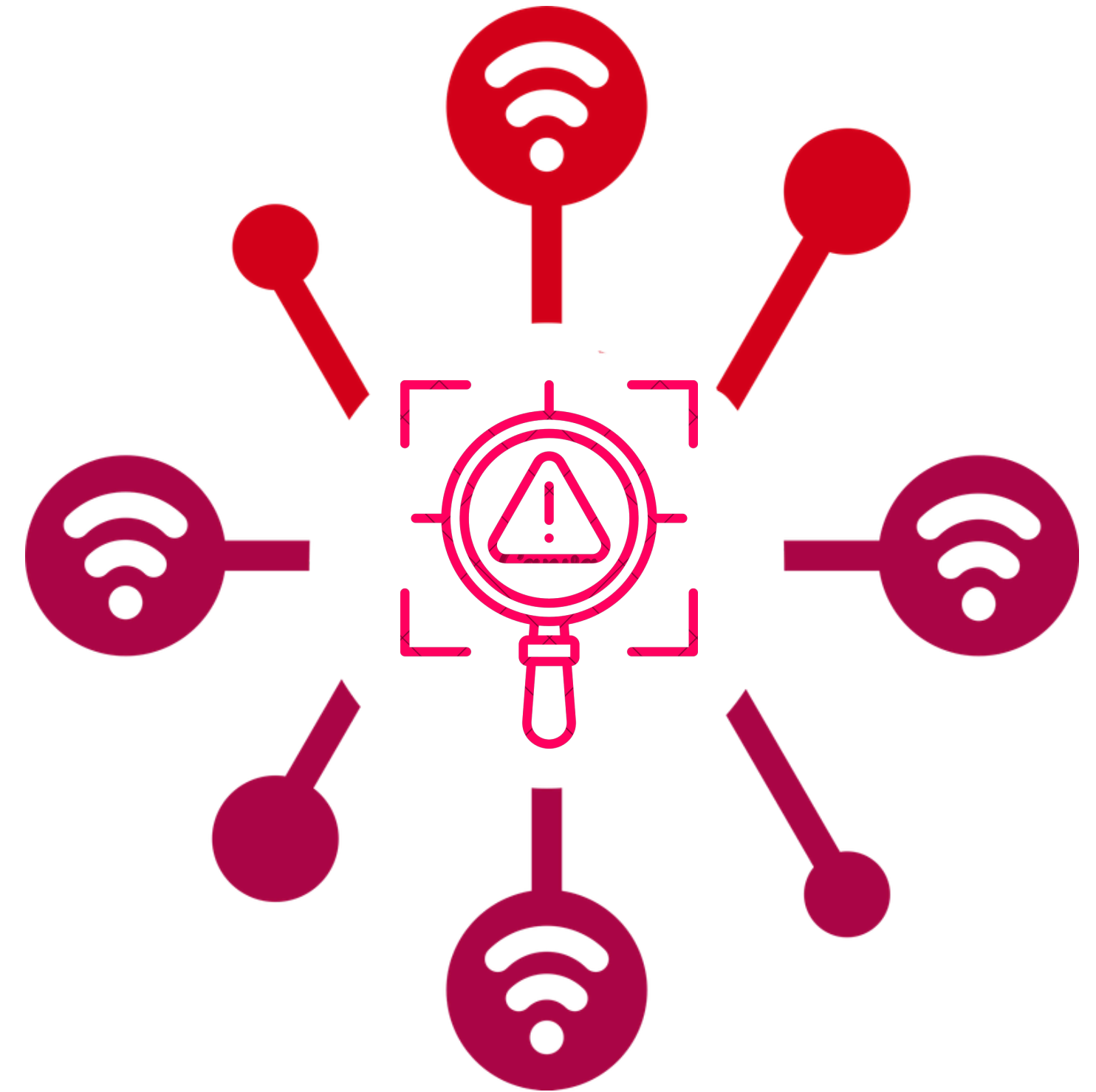


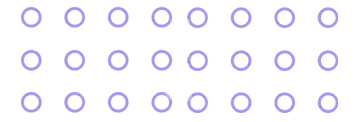
COMPREHENSIVE PERSPECTIVES ON INTRUSION DETECTION SYSTEMS IN MODERN CYBERSECURITY

Presented By :

Oussama Ben Sassi



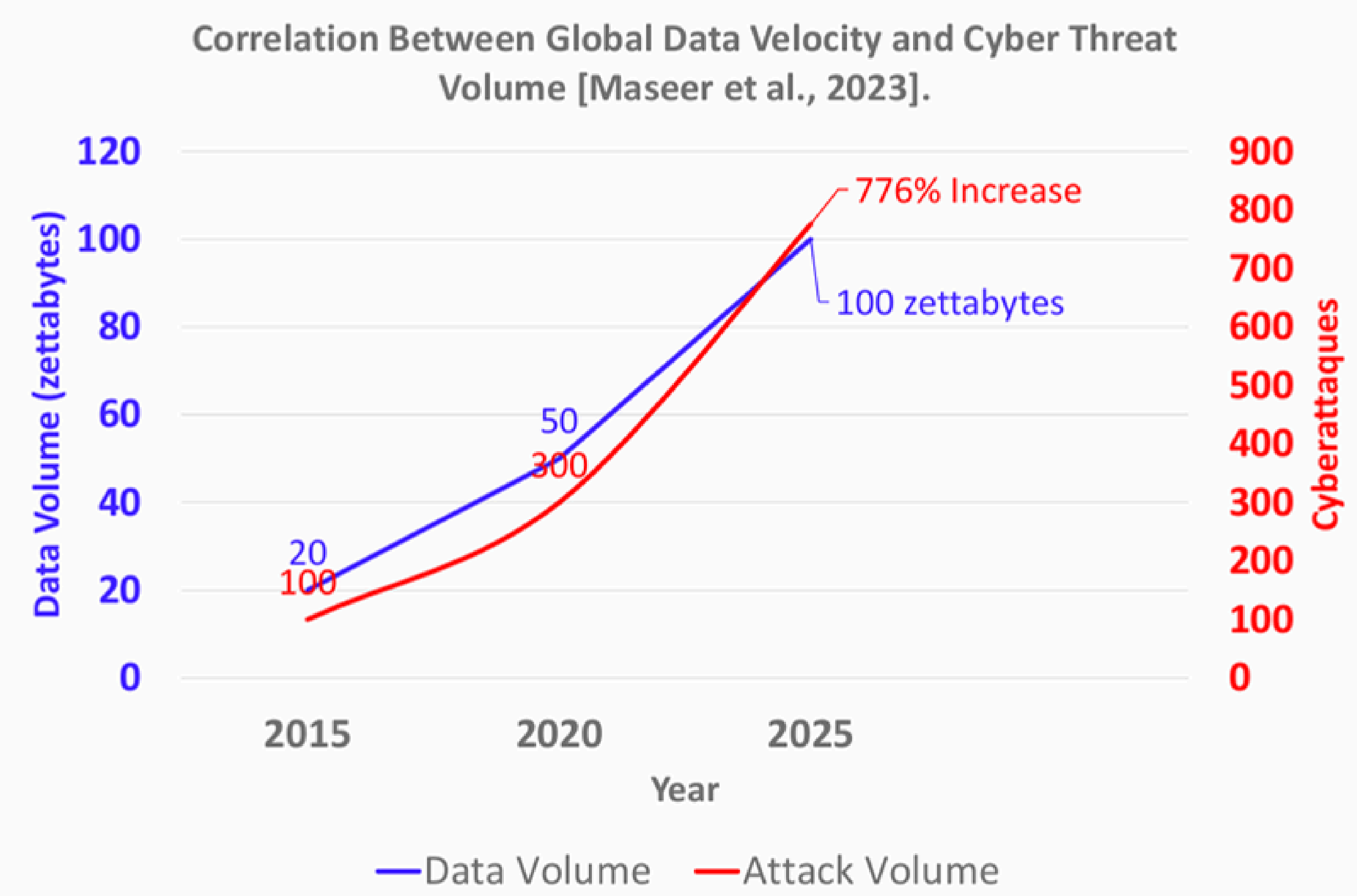
PRESENTATION OUTLINE



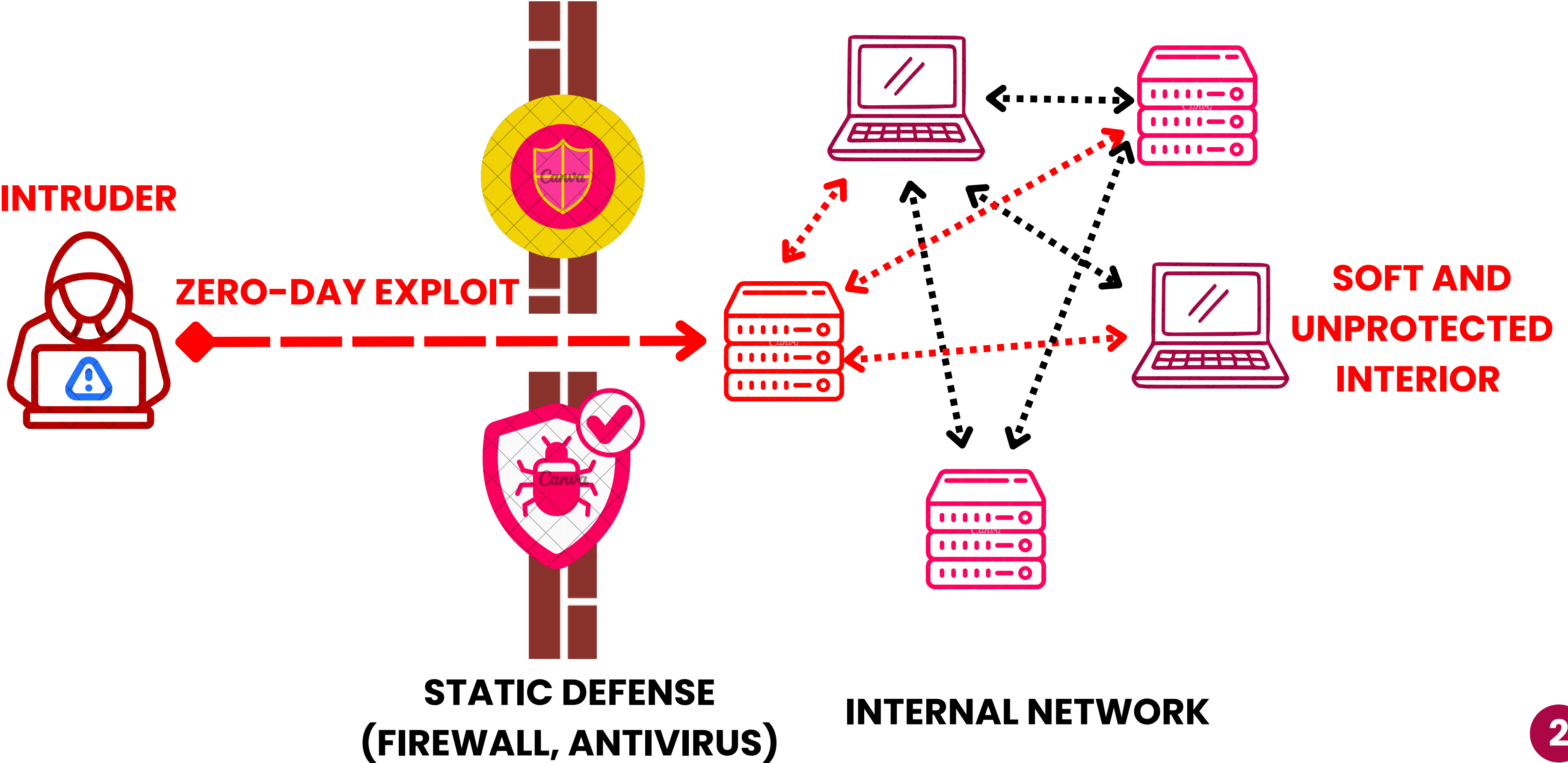
- 01 CURRENT CYBERSECURITY THREAT LANDSCAPE AND LIMITATIONS OF TRADITIONAL DEFENSES**
- 02 INTRUSION DETECTION SYSTEM (IDS) : CORE SECURITY PRINCIPLES, DEFENSE IN DEPTH AND THE PLACEMENT OF IDS IN THE NETWORK**
- 03 CLASSICAL AND INTELLIGENT DETECTION METHODS AND THEIR CHALLENGES.**
- 04 STATE-OF-THE-ART IDS DATASETS AND LIMITATIONS OF IDS BENCHMARKS**
- 05 DATA PREPROCESSING TECHNIQUES**
- 06 EVALUATION METRICS**
- 07 CONCLUSION**



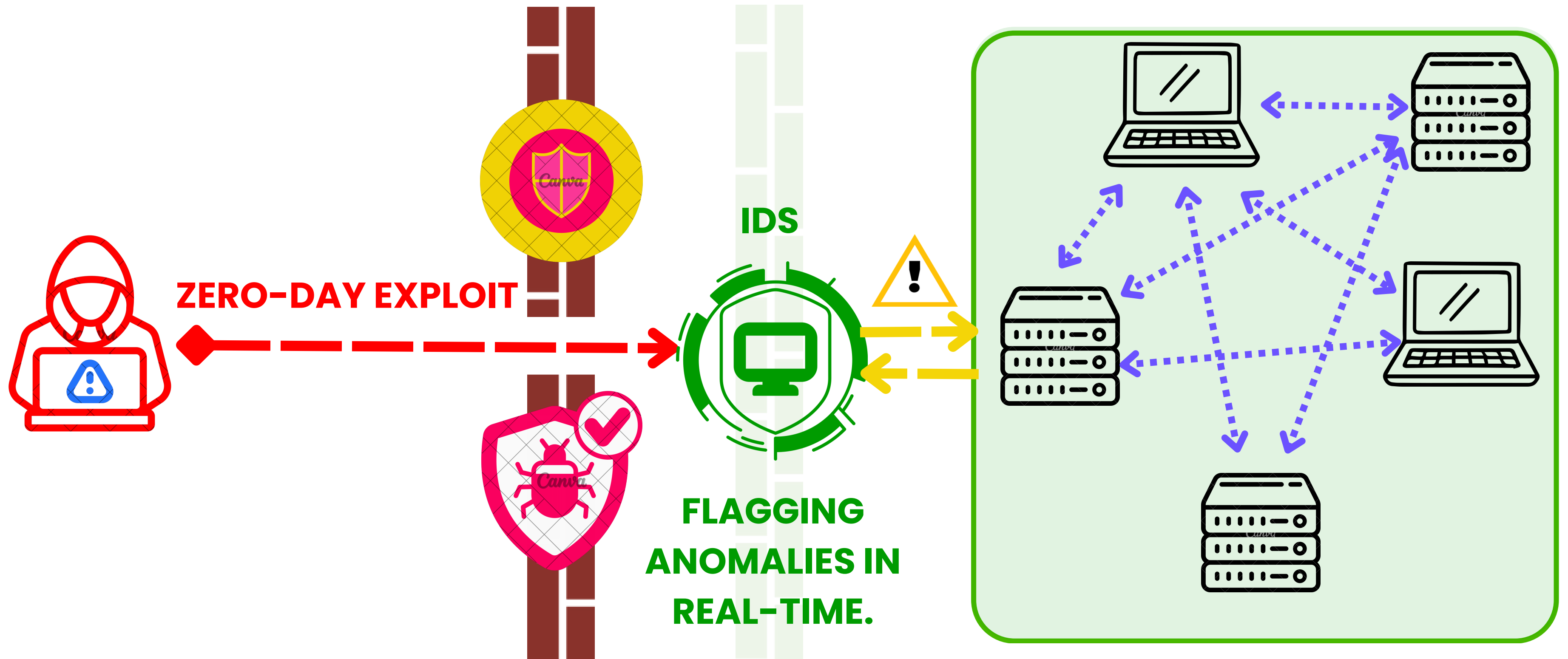
CURRENT CYBERSECURITY THREAT LANDSCAPE



LIMITATIONS OF TRADITIONAL DEFENSES

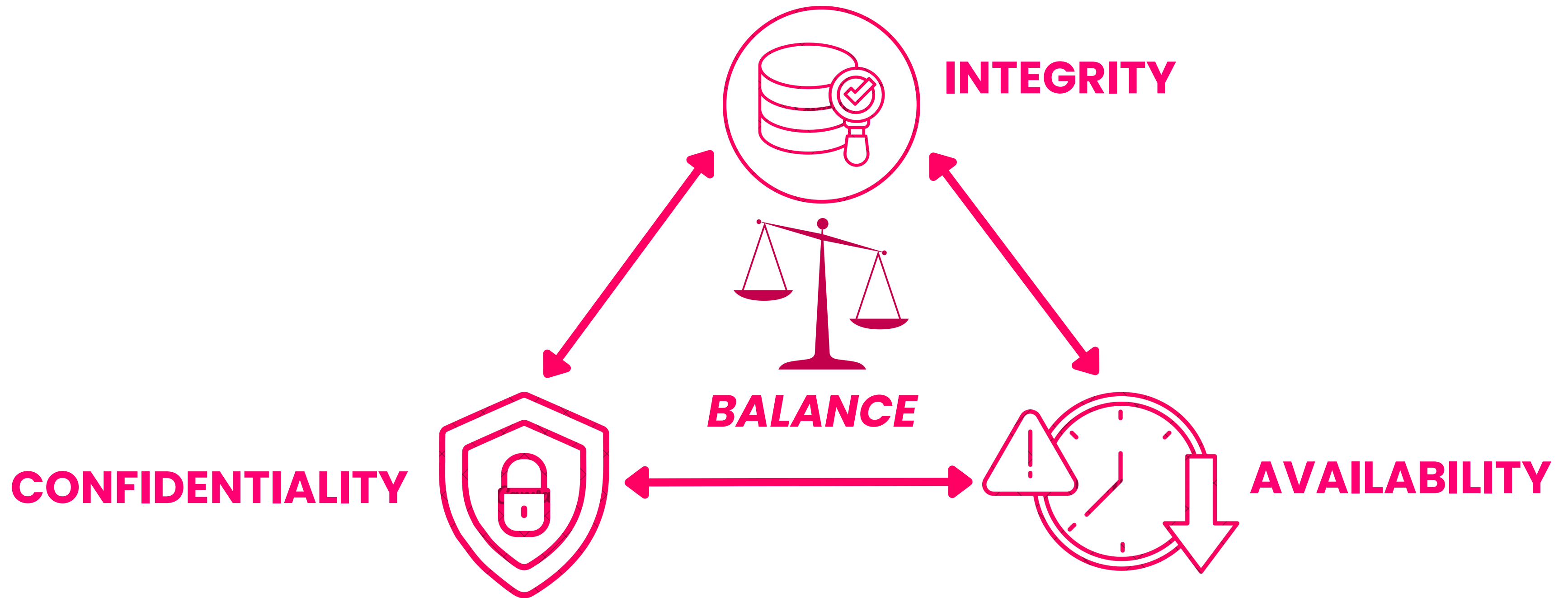


INTRUSION DETECTION SYSTEM (IDS)



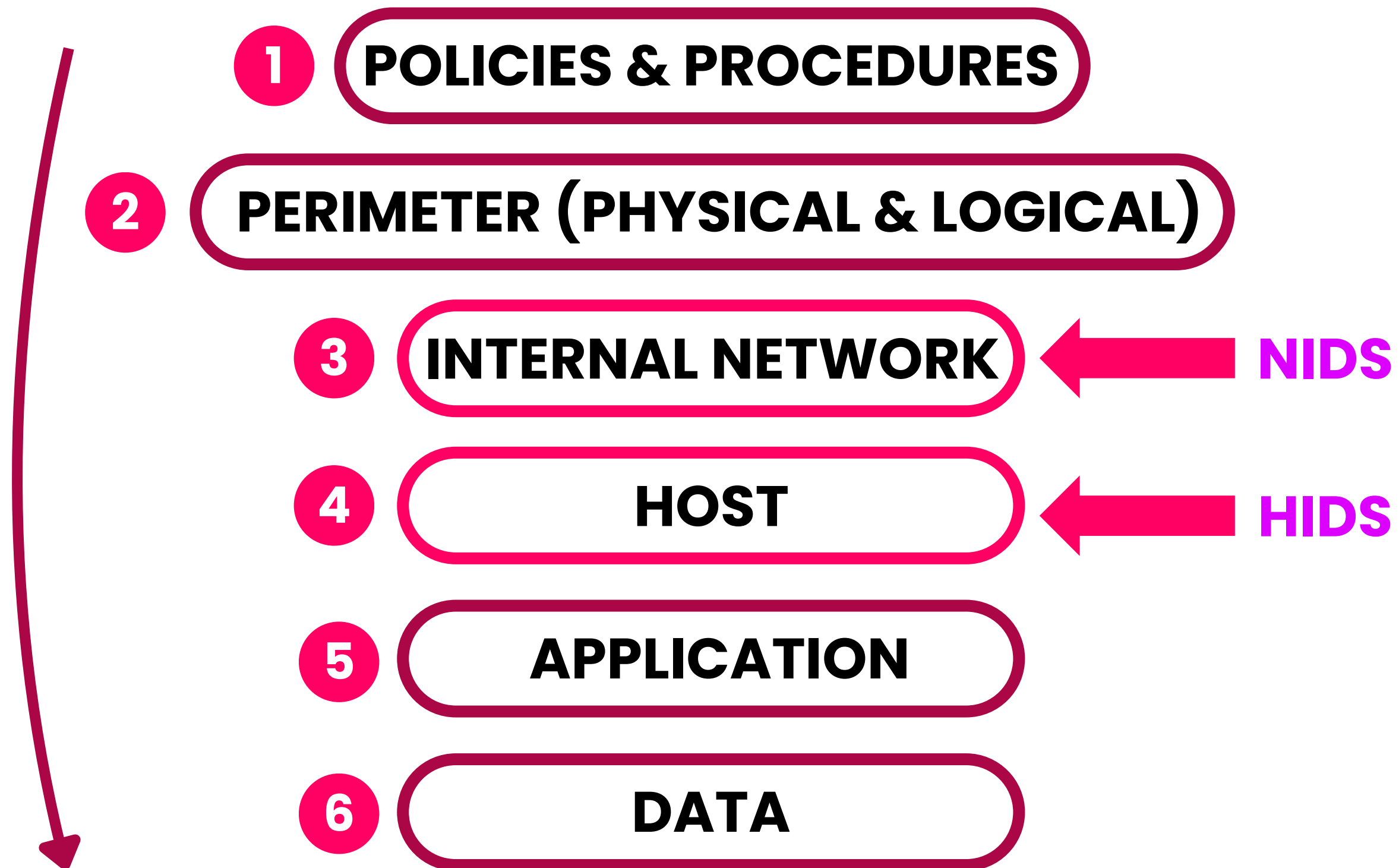
IDS CORE SECURITY PRINCIPLES

IDS RELY ON THE CIA TRIAD TO PROTECT INTERNAL NETWORKS
[ALI ET AL., 2024].

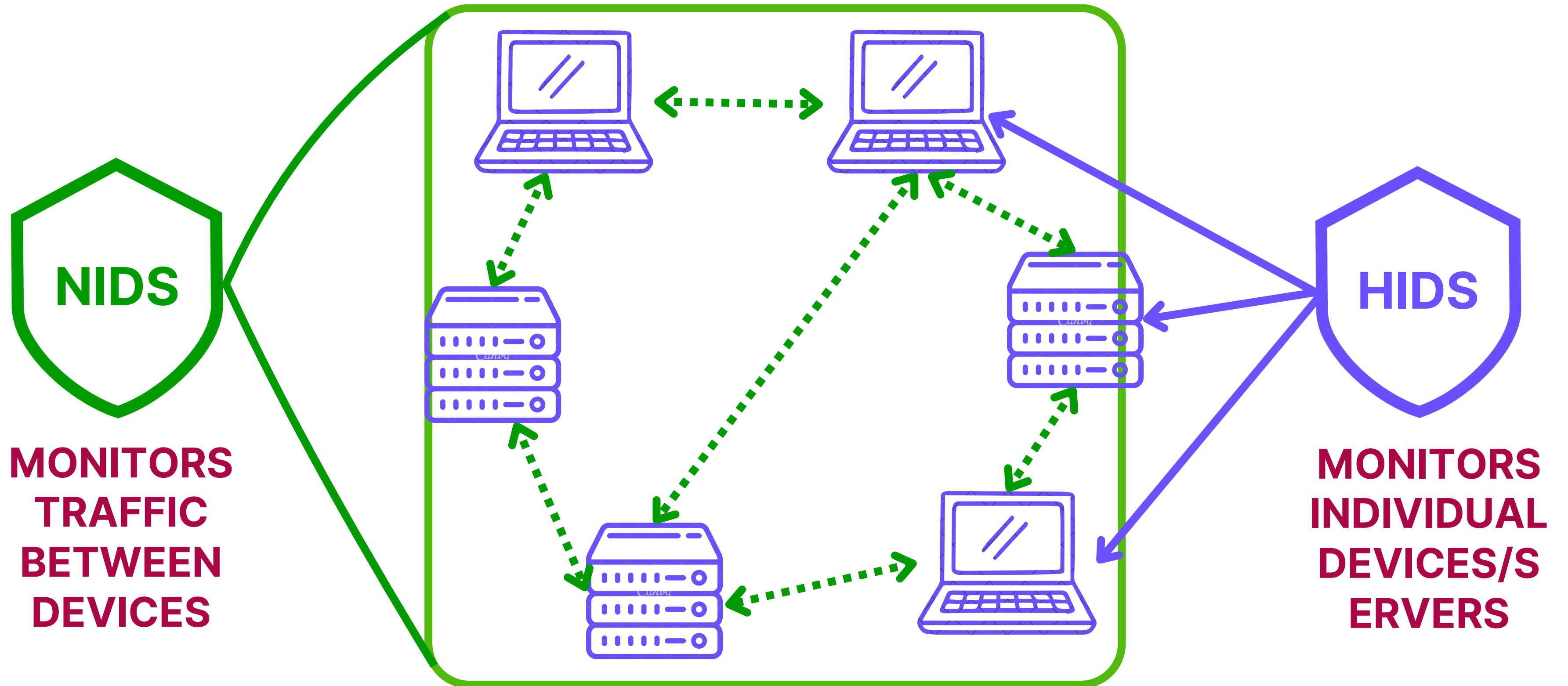


DEFENSE IN DEPTH

THE CIA TRIAD IS THE OBJECTIVE, **DEFENSE IN DEPTH** IS THE STRATEGY TO IMPLEMENT IT.

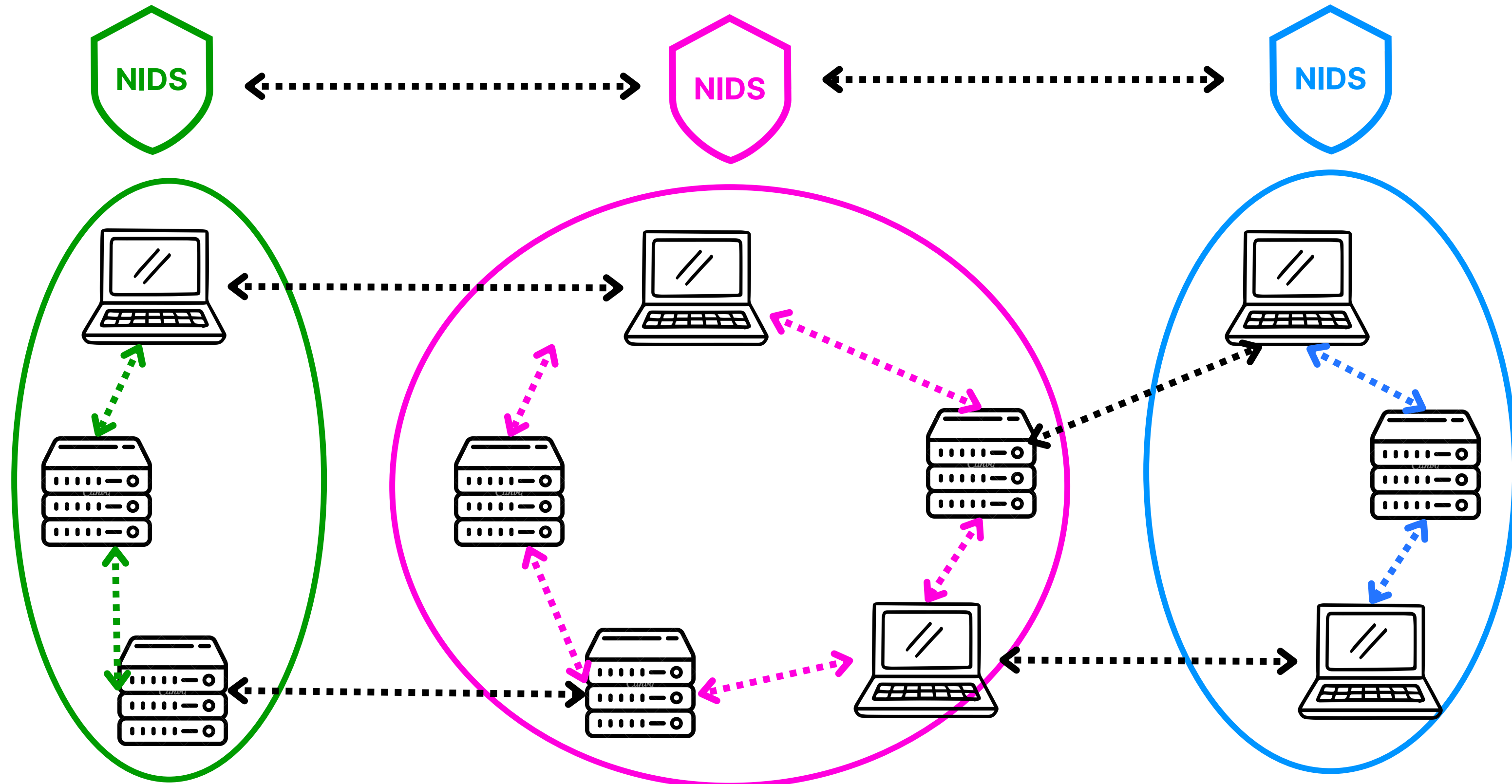


PLACEMENT OF IDS IN THE NETWORK



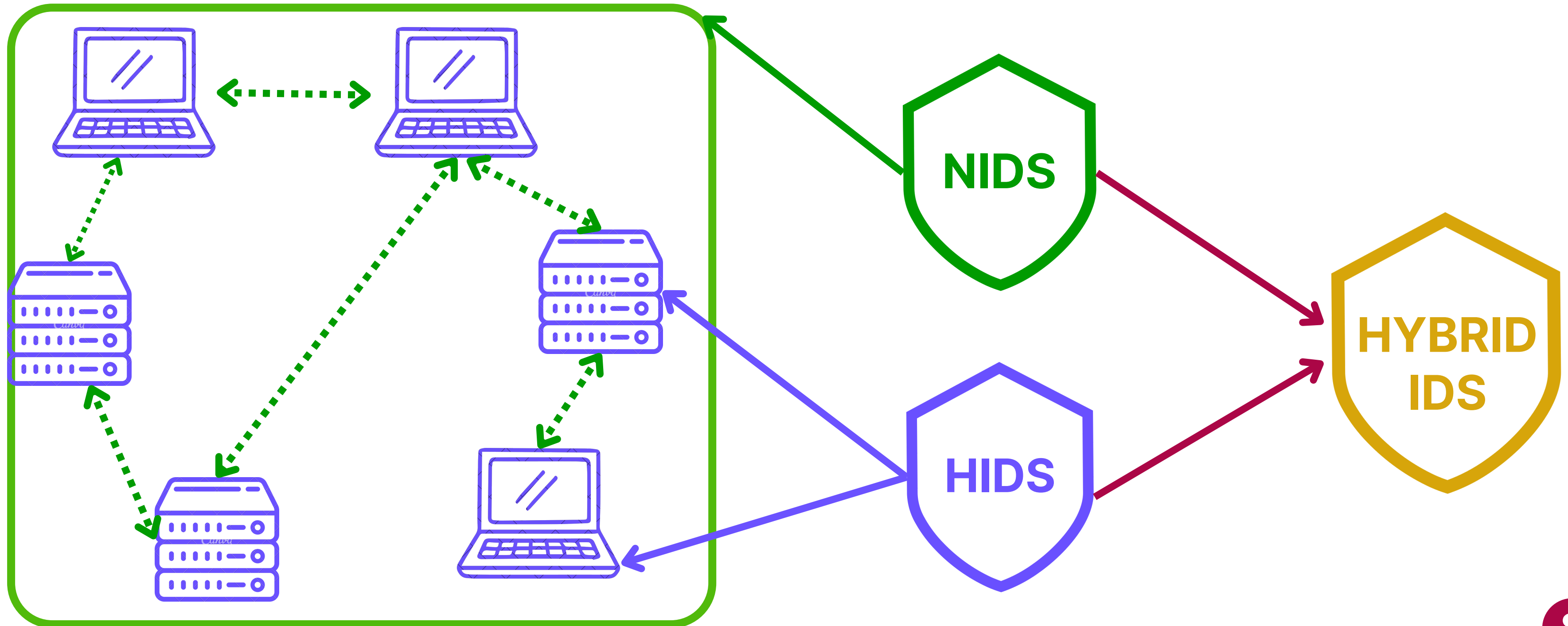
ADVANCED IDS ARCHITECTURES

MODERN ENVIRONMENTS ARE TOO LARGE FOR ISOLATED TOOLS. **DISTRIBUTED** AND **HYBRID** ARCHITECTURES ARE REQUIRED.

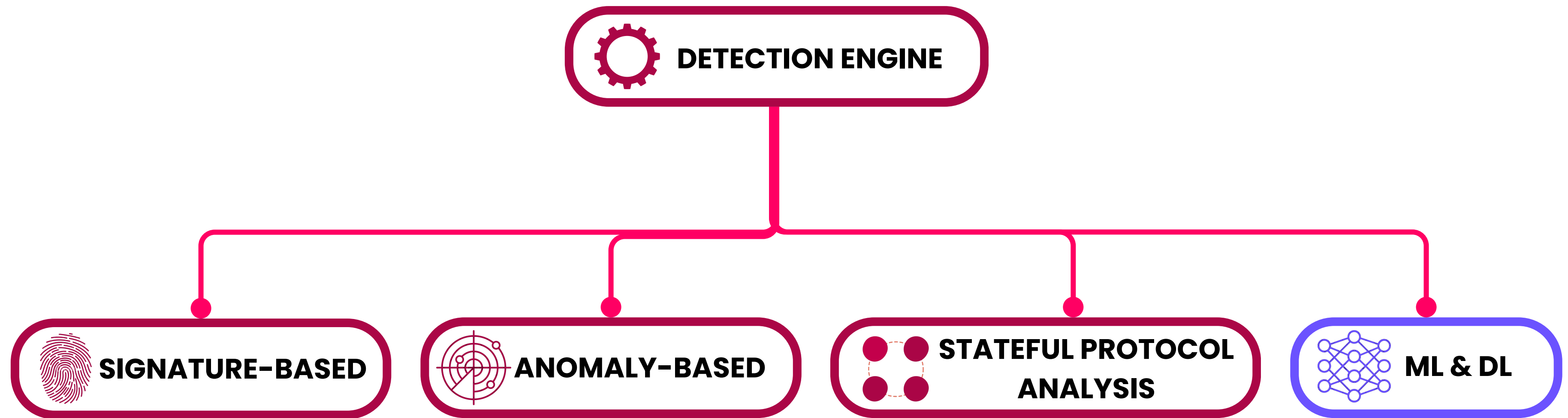


ADVANCED IDS ARCHITECTURES

HYBRID IDS: INTEGRATION OF NIDS AND HIDS DATA TO COMPENSATE FOR **INDIVIDUAL WEAKNESSES.**



DETECTION APPROACHES



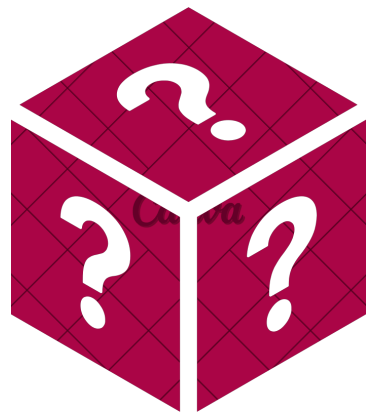
CLASSICAL DETECTION METHODS

Method	Logic	Strength	Weakness
Signature	Pattern matching	Very Fast	Can't detect unknown attacks
Anomaly	Normal vs. Deviation	Finds unknown attacks	High false positives
Stateful	Tracks the state and behavior of communication sessions	Precision	Slow in high-traffic environments

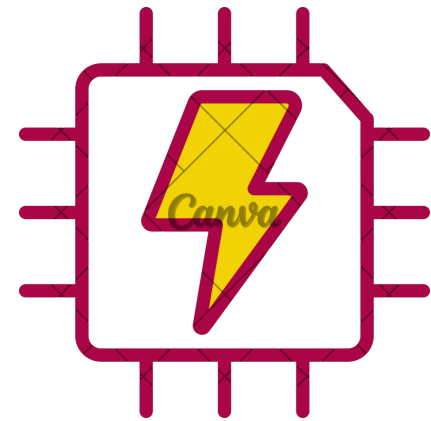
INTELLIGENT DETECTION (ML & DL)

Architecture	Principle	Benchmarks
LSTM [Jacob et al., 2024]	Sequential,long-term dependencies	97 %
CNN [Abdulganiyu et al., 2023]	Convolution, Automatic feature extraction	94 %
Random Forest [Jacob et al., 2024]	Ensemble of decision trees	99.51 %
Stacked Learning [Mamatha et al., 2025]	Layered model hierarchy	99 %
Hybrid CNN + AE [Abdulganiyu et al., 2023]	Spatial features + dimensionality reduction	~100%

CHALLENGES IN ML & DL APPROACHES.



**LACK OF
INTERPRETABILITY**
(XAI)



**RESOURCE
INTENSITY**
(TRAINING TIME,
INFERENCE..)



DATA IMBALANCE



**ATTACKS EVOLVE
OVER TIME**

STATE-OF-THE-ART IDS DATASETS

THE FOUNDATIONS

- ◆ **DARPA, 1998, KDD CUP, 1999 : SYNTHETIC TRAFFIC, HEAVY REDUNDANCY.**
- ◆ **NSL-KDD, 2009: REDUCED REDUNDANCY.**

MODERN

- ◆ **UNSW-NB15 , 2015 : HYBRID + SYNTHETIC**
- ◆ **CICIDS2017 : REALISTIC PACKETS (DDOS, BRUTE FORCE, BOTNET, WEB ATTACKS, INFILTRATION.**

THE FRONTIERS

- ◆ **CIC-IOT-2023, CIC-IOMT-2024 : IOT DATASET, MULTI-PROTOCOL TRAFFIC, 33 ATTACKS IN 7 CATEGORIES**

LIMITATIONS OF CURRENT IDS BENCHMARKS



KDD and NSL-KDD, contain **duplicate records. This leads models to simply memorize the data rather than learn to detect actual threats**
[Jacob et al., 2025]. (overfitting)



The vast majority of models (around 90%) are **never tested in a real-world environment.**
[Satılmış et al., 2024]. (complexity, concept drift)



Modern attackers use to blend in with **encrypted traffic**
[Ahmad et al., 2023]. (observable behavior)

DATA PREPROCESSING PIPELINE

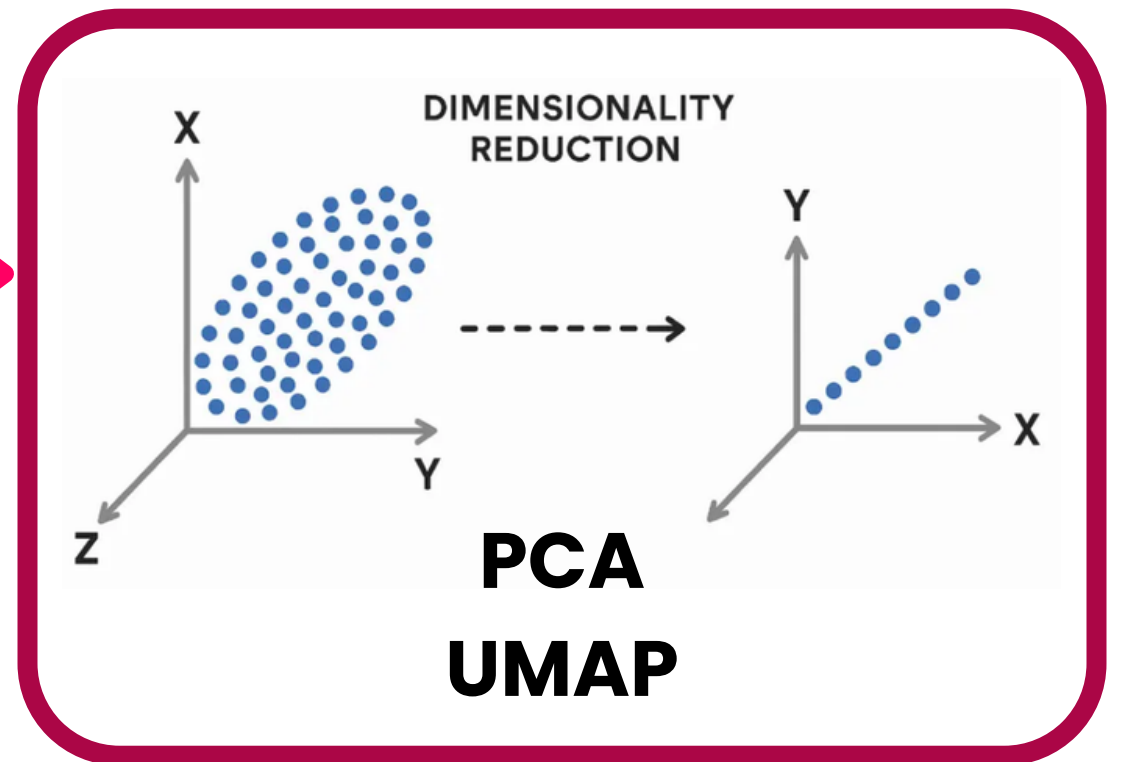
DATA CLEANSING

MIN-MAX (0-1)
Z-SCORE
QUANTILE
TRANSFORMER

REDUNDANCY MITIGATION

DELETION OF
DUPLICATE
ROWS/ATTRIBUTES.

DIMENSIONALITY REDUCTION



BALANCING

CLASS-WEIGHTS
SMOTE

MODEL

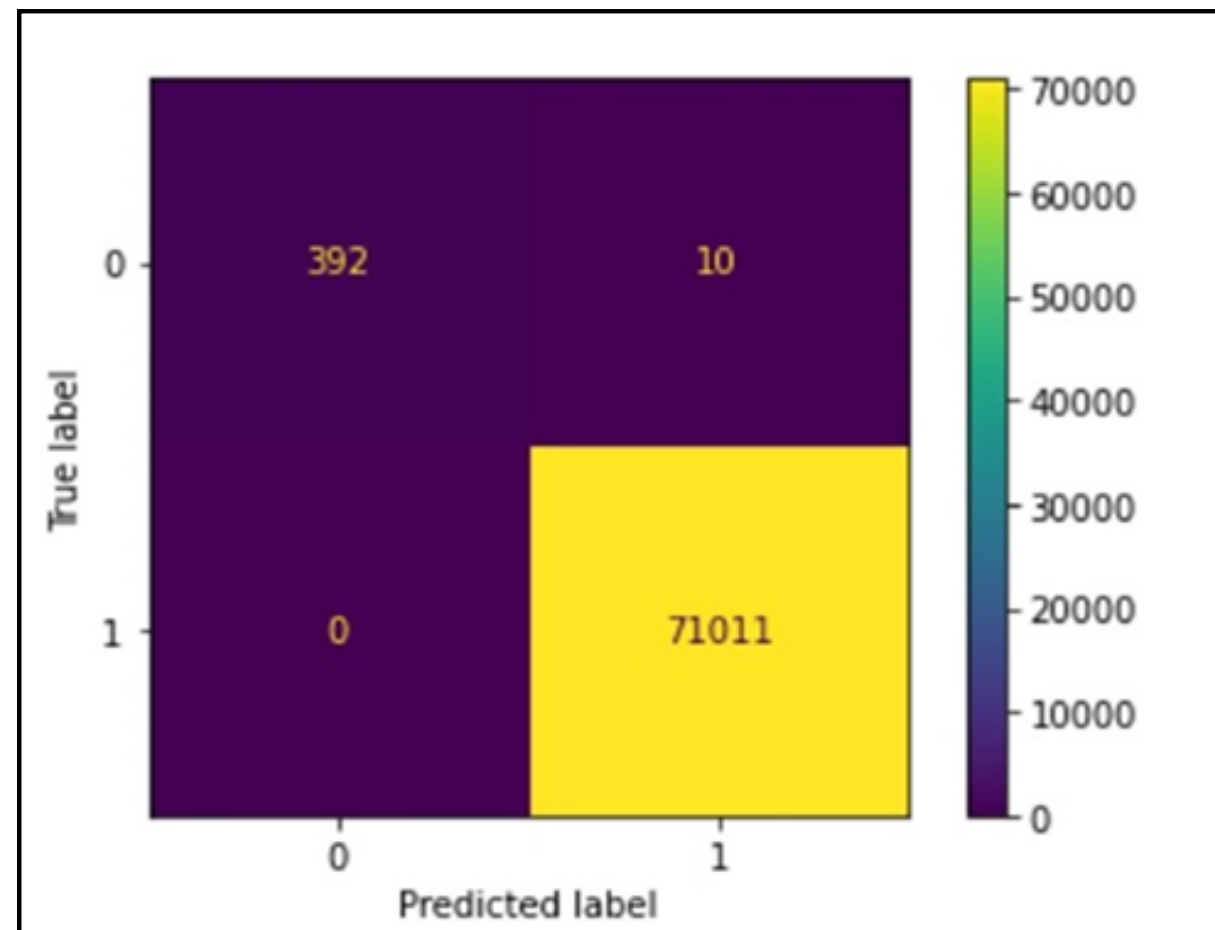
PROCESSED DATASET

PERFORMANCE
EVALUATION

EVALUATION OF INTRUSION DETECTION SYSTEMS

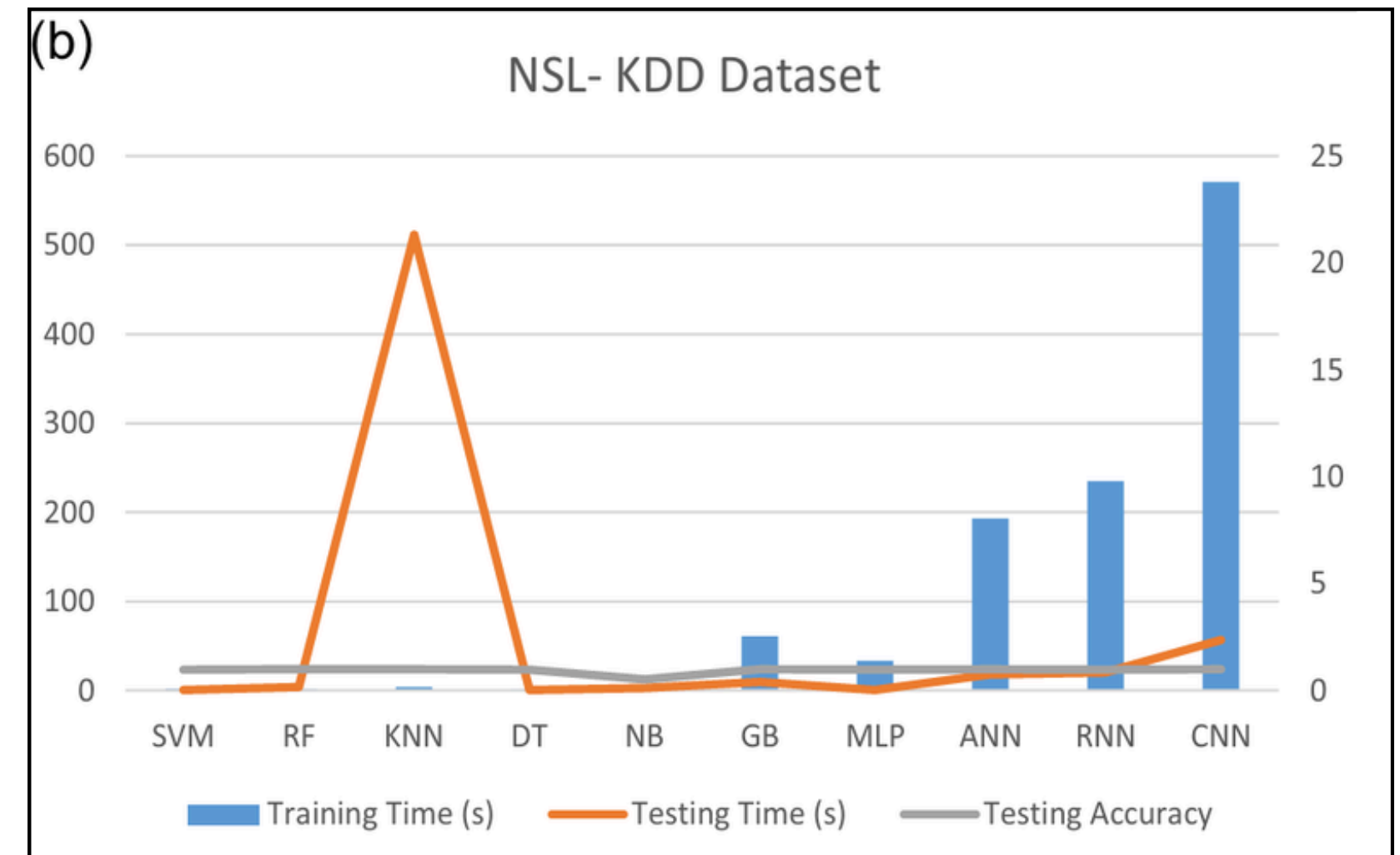
CLASSIFICATION QUALITY

- Confusion Matrix (TP, FP, TN, FN)
- **Accuracy**
- Precision $TP / (TP + FP)$
- Recall $TP / (TP + FN)$



OPERATIONAL PERFORMANCE

- inference time
- Resource Footprint
- Detection Latency

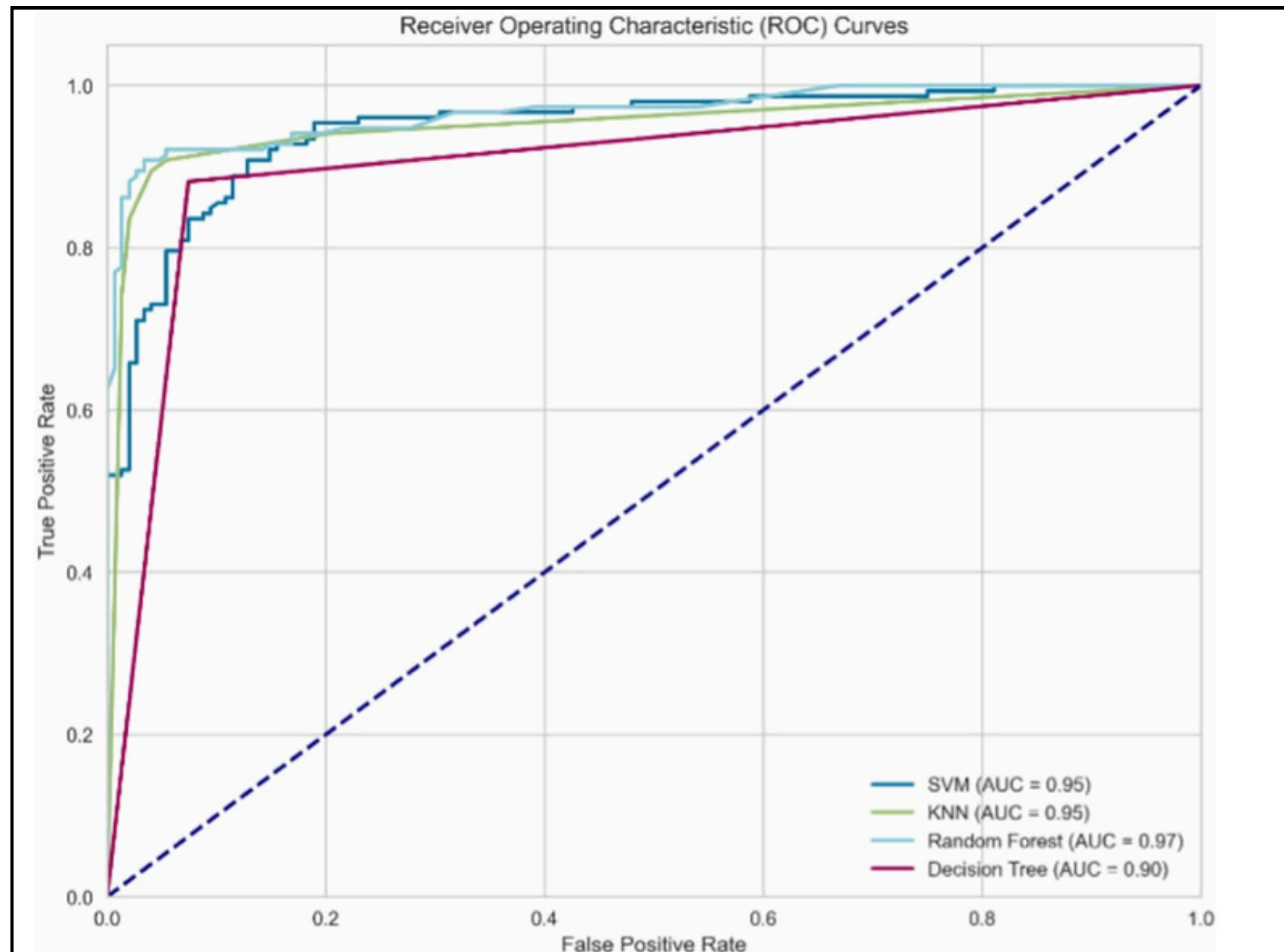


[Jacob et al., 2025]

EVALUATION OF INTRUSION DETECTION SYSTEMS

ERROR ANALYSIS

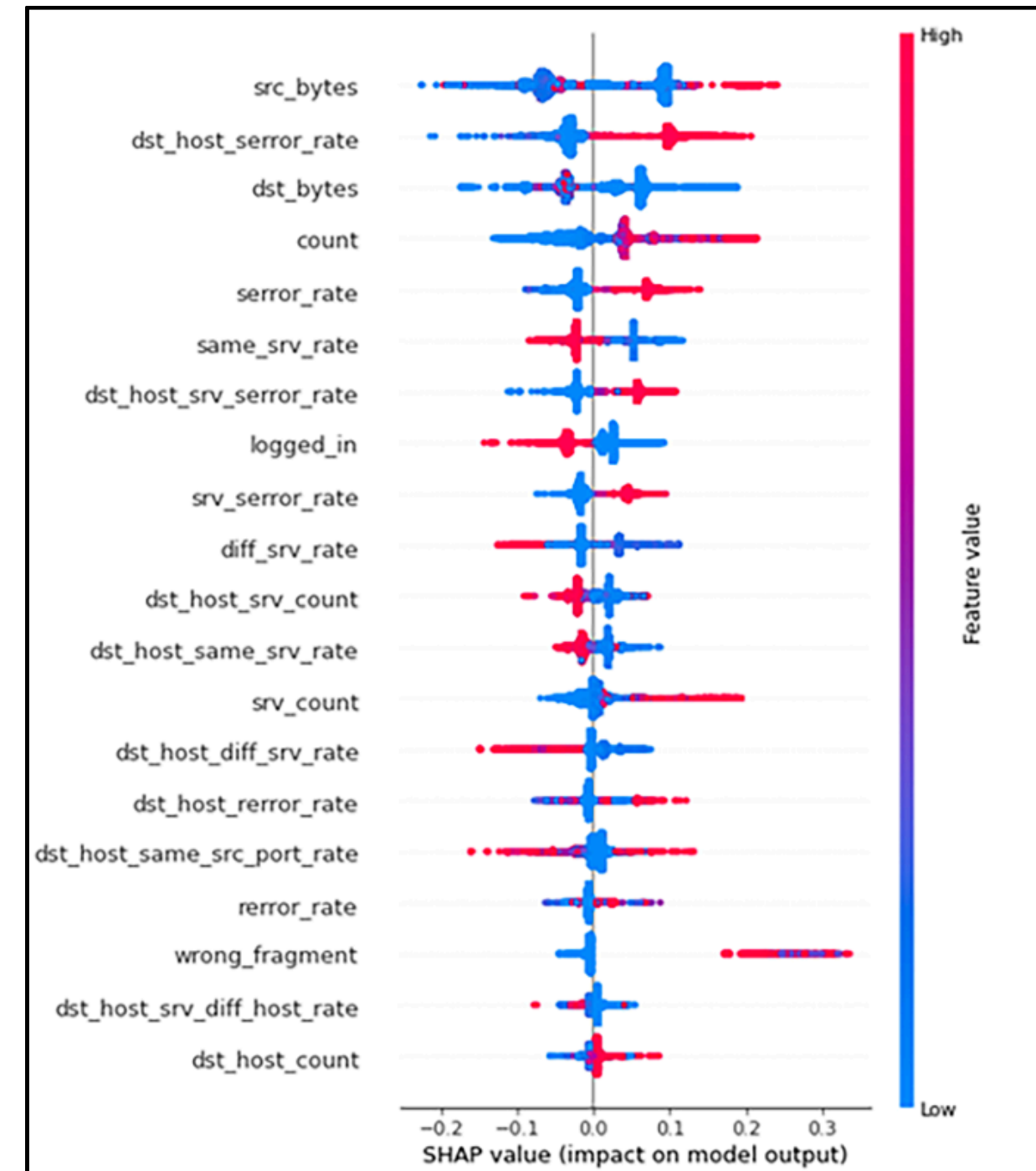
● ROC curves (Receiver Operating Characteristic)



[Ahmed et al., 2025]

EXPLAINABILITY (XAI)

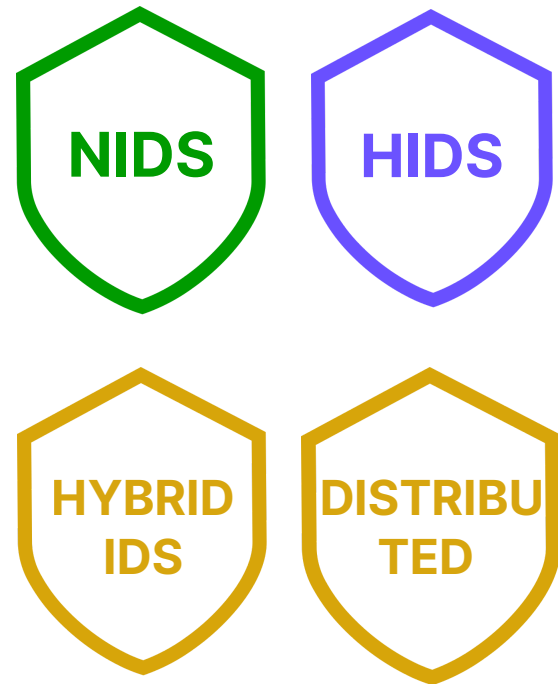
● SHAP (SHapley Additive exPlanations)



[Ben Ncir et al., 2024]

CONCLUSION

IDS



EVOLUTION

STATIC IDS



**ML AND BEHAVIORAL
MODELING**

DATASETS

NSL-KDD, 2009



UNSW-NB15, 2015



CIC-IOT2023

LIMITATIONS

DUPLICATE RECORDS

CLASS IMBALANCE

DATA CLEANSING

DIMENSIONALITY

REDUCTION

SMOTE

EVALUATION METRICS

CLASSIFICATION

OPERATIONAL

ERROR

EXPLAINABILITY

THANK YOU FOR YOUR TIME AND ATTENTION.

REFERENCES :

- Z. K. **MASEER**, R. YUSOF, B. AL-BANDER, A. SAIF, AND Q. K. KADHIM, "SYSTEMATIC REVIEW FOR ANOMALY NETWORK INTRUSION DETECTION SYSTEMS: DETECTION METHODS, DATASET, VALIDATION METHODOLOGY, AND CHALLENGES".
- A. H. **ALI** ET AL., "UNVEILING MACHINE LEARNING STRATEGIES AND CONSIDERATIONS IN INTRUSION DETECTION SYSTEMS: A COMPREHENSIVE SURVEY," FRONT. COMPUT. SCI., VOL. 6, P. 1387354, JUN. 2024, DOI: [10.3389/fcomp.2024.1387354](https://doi.org/10.3389/fcomp.2024.1387354).
- S. L. **JACOB** AND P. SULTANA HABIBULLAH, "A SYSTEMATIC ANALYSIS AND REVIEW ON INTRUSION DETECTION SYSTEMS USING MACHINE LEARNING AND DEEP LEARNING ALGORITHMS," JCCE, JUL. 2024, DOI: [10.47852/BONVIEWJCCE42023249](https://doi.org/10.47852/BONVIEWJCCE42023249).
- O. H. **ABDULGANIYU**, T. AIT TCHAKOUCT, AND Y. K. SAHEED, "A SYSTEMATIC LITERATURE REVIEW FOR NETWORK INTRUSION DETECTION SYSTEM (IDS)," INT. J. INF. SECUR., VOL. 22, NO. 5, PP. 1125–1162, OCT. 2023, DOI: [10.1007/s10207-023-00682-2](https://doi.org/10.1007/s10207-023-00682-2).
- P. **MAMATHA**, S. BALAJI, AND S. S. ANURAGHAV, "DEVELOPMENT OF HYBRID INTRUSION DETECTION SYSTEM LEVERAGING ENSEMBLE STACKED FEATURE SELECTORS AND LEARNING CLASSIFIERS TO MITIGATE THE DOS ATTACKS," INT J COMPUT INTELL SYST, VOL. 18, NO. 1, P. 20, FEB. 2025, DOI: [10.1007/s44196-025-00750-6](https://doi.org/10.1007/s44196-025-00750-6).
- H. **SATILMIŞ**, S. AKLEYLEK, AND Z. Y. TOK, "A SYSTEMATIC LITERATURE REVIEW ON HOST-BASED INTRUSION DETECTION SYSTEMS," IEEE ACCESS, VOL. 12, PP. 27237–27266, 2024, DOI: [10.1109/ACCESS.2024.3367004](https://doi.org/10.1109/ACCESS.2024.3367004).
- R. **AHMAD**, I. ALSMADI, W. ALHAMDANI, AND L. TAWALBEH, "ZERO-DAY ATTACK DETECTION: A SYSTEMATIC LITERATURE REVIEW," ARTIF INTELL REV, VOL. 56, NO. 10, PP. 10733–10811, OCT. 2023, DOI: [10.1007/s10462-023-10437-z](https://doi.org/10.1007/s10462-023-10437-z).
- U. **AHMED** ET AL., "SIGNATURE-BASED INTRUSION DETECTION USING MACHINE LEARNING AND DEEP LEARNING APPROACHES EMPOWERED WITH FUZZY CLUSTERING," SCI REP, VOL. 15, NO. 1, P. 1726, JAN. 2025, DOI: [10.1038/s41598-025-85866-7](https://doi.org/10.1038/s41598-025-85866-7).
- C. E. **BEN NCIR**, M. A. BEN HAJKACEM, AND M. ALATTAS, "ENHANCING INTRUSION DETECTION PERFORMANCE USING EXPLAINABLE ENSEMBLE DEEP LEARNING," PEERJ COMPUTER SCIENCE, VOL. 10, P. E2289, SEP. 2024, DOI: [10.7717/PEERJ-CS.2289](https://doi.org/10.7717/PEERJ-CS.2289).